Tô Gia Bảo

AML Analyst

Thu Duc, Ho Chi Minh City, Vietnam

+84 96 196 4353 | baokdl2226@gmail.com | https://tgblab.io.vn | ToGiaBaoKDL | togiabao

Summary

Data Engineer in digital banking with hands-on experience supporting anti-money laundering (AML) and fraud detection through transaction monitoring and behavioral risk analytics. Developed 40+ behavioral features and a 10-signal scoring engine to help Compliance and Fraud Operations prioritize account reviews. Strong Python, SQL, Athena, and dbt skills, with a focus on reliable data, detection rules, and alert quality.

Experience

Timo Digital Bank by BVBank — *Ho Chi Minh City, Vietnam*

Data Engineer — *Mar. 2025 – Present*

- Engineered a daily incremental AML and fraud feature mart on AWS Athena and dbt, tracking 40+ behavioral features across millions of transactions to identify fund pass-through, structuring, and unusual transaction arrival patterns.
- Developed a 10-signal risk scoring engine to identify potential mule accounts through transaction velocity, nighttime bursts, small incoming transfers, and concentrated cash-outs; added balance retention gates and two-tier review queues to reduce false-positive triage workload for Compliance and Fraud Operations by approximately 50% while retaining confirmed mule cases for review.
- Built daily alerts comparing intraday and month-to-date transaction frequency and gross transaction value against a three-month historical baseline, flagging sudden behavioral spikes for further review.
- Implemented MariaDB ingestion workflows with Amazon EMR and AWS Glue, applying PII masking in the curated layer and dbt transformations to deliver analytics-ready datasets.
- Delivered Tracking247: a Streamlit app accepting shipment IDs from the Customer team, with Airflow syncing shipment status into MongoDB.
- Migrated Timo Metrics from Loki/Grafana to SigNoz, centralizing logs and metrics for operational monitoring and troubleshooting.
- Developed a FastMCP server for conversational analytics in Slack, using dbt docs, Lightdash dashboards, and semantic metadata to support self-service data discovery.

Projects

Mini Lakehouse — *GitHub*

Lakehouse & Platform Engineering — *Jul. 2026 – Present*

- Built a contract-driven AWS lakehouse on S3, Iceberg, Glue, Athena, and EMR Serverless, with Airflow and platform services self-hosted on OCI and infrastructure managed through Terraform.
- Delivered batch and streaming pipelines for GitHub activity, arXiv research, and SSI market data, with dbt transformations, Lightdash and Sigma BI dashboards, Modal GPU OCR, and Netdata monitoring via Cloudflare Zero Trust.

VN Market Pulse — *GitHub*

AI Research & Content Platform — *Jul. 2026*

- Built a Vietnamese research pipeline with multi-source search, deduplication, and reranking to generate source-backed posts.
- Used PydanticAI for typed output and validation, with retrieval budgets, retries, usage tracking, and CLI/Streamlit interfaces.

Skills

AML & Fraud Analytics	Transaction monitoring, mule account detection, behavioral risk scoring, alert prioritization
Programming & Analytics	Python, SQL, Amazon Athena, Lightdash, Sigma
Pipelines & Modeling	dbt, Airflow, Spark/PySpark, dimensional modeling
Data Quality & Testing	dbt tests, Great Expectations, pytest
Data Platforms & Tools	Amazon S3, EMR, AWS Glue, PostgreSQL, MariaDB, Git

Education

University of Science, VNU-HCM (HCMUS) — *Ho Chi Minh City, Vietnam*

Bachelor of Data Science | GPA: 3.78/4.00 — *2022 – 2026*

Awards & Certifications

2025 **Top 25**, AIC Challenge
2025 **Recipient**, Ninety Eight Scholarship
2024 **Top 10**, RMIT Data Analytics Competition
2024 **Google Business Intelligence**, Professional Certificate
2024 **Google Data Analytics**, Professional Certificate

Publication

GalaxyAssistant: An Intelligent Assistant for Multimedia Event Retrieval — *SOICT*

Co-author — *2025*

Languages

English TOEIC Listening & Reading: 910 | TOEIC Speaking & Writing: 350